Spin-0, Spin-1 and Spin-2 Deformations of $S^3 \times S^1$ in Einstein–Cartan–Nieh–Yan Theory

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Abstract

We organize spatially homogeneous deformations of $S^3 \times S^1$ in Einstein–Cartan theory with constant complete Nieh–Yan coupling into spin-0, spin-1 and spin-2 representations. A closed static potential determines radius–torsion couplings, a conditional radius minimum, positive twist stiffness and volume-preserving shape dependence. The spin-2 sector has a continuous bifurcation of constrained static minima at $\epsilon_c = 1/\sqrt{2} - 1$, a square-root amplitude law and a second-derivative jump; minimizing the static potential over shape and twist at fixed scalars selects the round, zero-twist geometry. We also derive the exact joint action for periodic circle-dependent scalar, shape and twist profiles. Its scalar and vector restrictions recover the radius derivative term and the positive amplitude-quartic twist term $|\omega \times \omega'|^2$, while the zero-twist shape restriction has a positive tracefree derivative term. The derivative quadratic form separates the sectors at a constant round, zero-twist reference. Nonlinear derivative cross terms and a shift coupling connect them. An explicit smooth periodic shape–twist family makes the joint Euclidean action unbounded below even at fixed scalar controls. Static minima and the global circle functional thus have distinct variational properties. Curvature identities, boundary currents and reproducible data complete the model description.

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1 Introduction

Homogeneous geometries provide a finite set of variables in which the dependence of a gravitational action on size, shape and torsion can be studied explicitly. In Einstein–Cartan (EC) theory the coframe and a metric-compatible connection allow these ingredients to be varied separately [3]. The complete Nieh–Yan (NY) term fixes how the torsion and curvature contributions enter the same differential-form identity [4]. On $S^3 \times S^1$, spatial rotations organize the geometric variables into scalar, vector and traceless-tensor representations.

Paper01 classified radius extrema on fixed homogeneous families, while Paper02 established curvature identities and shape bounds including an irreducible EC-Weyl correction [1, 2]. Here we study the EC member of this family, with the curvature-squared coefficient set to zero, and develop its spin-0, spin-1 and spin-2 geometric sectors. The term *spin* refers to the spatial $SO(3)$ representation of a deformation. We use it to organize variables and interactions; physical propagation requires a separate dynamical and constraint analysis.

The three sectors organize both a static potential and an exact circle-dependent action. Spin-0 gives a conditional radius minimum, radius–torsion couplings and a radial derivative term. Spin-1 has a positive static twist cost and, at fixed scalar controls and round shape, a finite-amplitude derivative term proportional to $|\omega \times \omega'|^2$. Spin-2 gives a continuous bifurcation of constrained static minima, a square-root amplitude law and a second-derivative jump; minimizing the static potential over constant volume-preserving shapes and twist at fixed scalars selects the round, zero-twist geometry. At zero twist and fixed radius, the circle variation supplies a nonnegative shape derivative term.

The joint circle action connects these results through derivative cross terms and the spatial shift. At the round, zero-twist reference, the derivative quadratic form separates the sectors. Nonlinear cross terms couple them, and an explicit periodic shape–twist family makes the joint Euclidean action unbounded below even at fixed scalar controls. Static shape minima and minima of the circle functional are consequently different variational questions. Section 2 gives the general action, Sections 3–5 develop the three sectors, and Section 6 derives the interactions and coupled family. Section 7 relates these results to Papers01/02.

2 Model and irreducible geometric variables

2.1 Coframe and action

Take spatial indices $i, j, k = 0, 1, 2$, circle index 3 and orientation $\epsilon^{0123} = +1$. The circle coordinate has period one. The left-invariant forms obey $d\sigma^i = 2\epsilon_{ijk}\sigma^j \wedge \sigma^k$, with the spatial pair summed. For $r, L > 0$ and a real matrix M of determinant one, define

$$\begin{aligned} e^i &= rM^i{}_j\sigma^j, & e^3 &= L(dy + \omega_i\sigma^i), \\ e'^a &= O^a{}_b e^b, \quad O \in SO(4), & g &= \delta_{ab}e'^a e'^b, \qquad \frac{d\mathcal{V}_4}{dy} = \frac{\pi^2 L r^3}{4}. \end{aligned} \tag{M1}$$

The spatially integrated four-volume density in (M1) integrates to $\pi^2 L r^3/4$ for constant r, L. The physical round radius is $r/2$. The three mixing parameters are $\delta_i = \tan\chi_i$ in $O = O_{03}(\chi_0)O_{13}(\chi_1)O_{23}(\chi_2)$. They rotate the coframe without changing the metric. Geometric torsion is specified in the resulting frame by

$$\begin{aligned} T_{ijk} &= \frac{2\eta}{r}\epsilon_{ijk}, \\ T^{(V)}_{abc} &= \frac{1}{3}(\delta_{ac}V_b - \delta_{ab}V_c), \qquad V_b = V\delta_{b3}, \\ \mathcal{S} &= -\int \mathrm{vol}_4 \left(\frac{R}{2\kappa^2} + \theta N\right), \qquad N\,\mathrm{vol}_4 = d(e'^a \wedge T_a). \end{aligned} \tag{M2}$$

The axial term is zero if any index is 3. AX means $V = 0$, VT means $\eta = 0$, and MX allows both. The coupling θ is constant, and no curvature-squared term is included. The connection is metric compatible and determined by this coframe and geometric torsion. Appendix A states its components; Appendix I gives the sign map to the code accompanying Papers01/02. Prescribing torsion components in the rotated frame is distinct from simultaneously rotating all fields as a gauge transformation.

Our sign convention extends the reduced-potential convention $U = -S$ of Papers01/02: on the common EC plus complete-NY model, $\mathcal{S} = -S_{01,\mathrm{FULL}} = -S_{02}|_{\alpha=0}$. Thus $\mathcal{S} = U$ on constant profiles. All lower-bound statements below concern U or $\mathcal{S}$ with this sign.

2.2 The common static potential

For constant $M, \omega, O, r, L, \eta, V$, set $S = MM^{\mathsf{T}}$. Cartan contraction gives the static action U, with F collecting its shape dependence:

$$\begin{aligned} U &= \mathcal{K}\left[F(S) + \frac{3\eta^2}{4} + \frac{V^2 r^2}{12} + \frac{L^2}{r^2}\|M\omega\|^2\right], \\ F(S) &= 2\mathrm{tr}(S^2) - [\mathrm{tr}(S)]^2, \qquad \mathcal{K} = \frac{\pi^2 L r}{\kappa^2} > 0. \end{aligned} \tag{M3}$$

Complete NY vanishes pointwise on this constant family, as proved in Section 6. The scalar action is independent of the mixing rotation: the torsion norms are invariant and their divergence vanishes. The coframe is global on $S^3 \times S^1$, so for smooth periodic circle variations the integral of complete NY also vanishes, although its local density can be nonzero.

2.3 Representation and variation families

The scalar variables are (r, L, η, V). Twist ω_i and the three mixing coordinates transform as spatial vectors near the round frame. A volume-preserving shape can be parametrized by $M = e^h$, with h symmetric and traceless; its five normalized components form spin-2. At an axial reference shape we use $M = M_0 e^h$ instead. Appendix C lists the basis. At the round point, rotation invariance separates different irreducible representations in the static quadratic expansion; scalar variables may still mix with one another.

Static extrema concern (M3) with designated coordinates varied at fixed controls. For the circle functional all coordinates may be smooth periodic functions of y, with $r, L > 0$ and $\det M = 1$. The scalar restriction sets $M = I, \omega = 0$; the vector restriction fixes r, L, η, V and sets $M = I$. Both follow from the joint action below. We keep their fixed data explicit when stating stationary or minimizing properties.

2.4 The joint circle action

Let $J = Lr^3$. The induced three-metric, lapse and shift in the invariant spatial basis are

$$\gamma = r^2 M^{\mathsf{T}} M + L^2 \omega\omega^{\mathsf{T}}, \qquad \mathcal{N}^2 = \frac{J^2}{\det\gamma}, \qquad \beta = L^2 \gamma^{-1}\omega. \tag{M4}$$

Define $[\beta]_\times v = \beta \times v$. The normalization of the spatial structure constants gives

$$\begin{aligned} &\Omega_\beta = 4[\beta]_\times, \quad \ell = \mathcal{L}_\beta \gamma = \gamma\Omega_\beta - \Omega_\beta\gamma, \\ &X = \gamma^{-1}\gamma', \qquad\qquad Y = \gamma^{-1}\ell, \qquad \mathrm{tr}Y = 0. \end{aligned} \tag{M5}$$

A prime denotes d/dy. The Lie derivative ℓ is generally nonzero for anisotropic shape. After integrating the total derivatives around the circle, the exact action is

$$\mathcal{S}_{\rm joint} = \int_0^1 dy\,[U + \mathcal{L}_\partial],$$
$$\mathcal{L}_\partial = \frac{\pi^2 J}{32\kappa^2 \mathcal{N}^2}\left[\mathrm{tr}(X^2) - (\mathrm{tr}X)^2 - 2\mathrm{tr}(XY)\right]. \tag{M6}$$

Here U is the static expression (M3) evaluated at the instantaneous coordinates. Its shift-dependent static contribution is already included, so no extra Y^2 term is added to (M6). The first two trace terms form the velocity quadratic form; the last is linear in velocities. Torsion derivatives and mixing rotations contribute only total derivatives, displayed in Appendix J. The constant complete NY coupling therefore does not enter the periodic bulk action. Formula (M6) holds at finite amplitude within this spatially homogeneous Euclidean family; it is not a fourth-order truncation.

3 Spin-0: radius, torsion and circle scale

3.1 Static couplings and radius minimum

At the round shape and zero twist the scalar potential is

$$U_0 = \frac{\pi^2 L r}{12\kappa^2}(V^2 r^2 + 9\eta^2 - 36). \tag{S1}$$

For fixed L, η, V, a strict positive-radius minimum exists exactly when $V \neq 0$ and $\eta^2 < 4$. Its position and curvature are

$$r_* = \frac{\sqrt{12 - 3\eta^2}}{|V|}, \qquad U_{rr}(r_*) = \frac{\pi^2 L V^2 r_*}{2\kappa^2} > 0. \tag{S2}$$

Thus $\eta = V = 1$ gives $r_* = 3$. At $V = 0$ the potential is linear in r: it decreases for $\eta^2 < 4$, is constant for $\eta^2 = 4$, and increases for $\eta^2 > 4$. If $V \neq 0$ and $\eta^2 \geq 4$, it increases for $r > 0$ and has no interior minimum. This classification distinguishes zero radial curvature from an actually flat potential. In particular, AX with $\eta = 1$ has $U_r = -9\pi^2 L/(4\kappa^2)$, despite $U_{rr} = 0$.

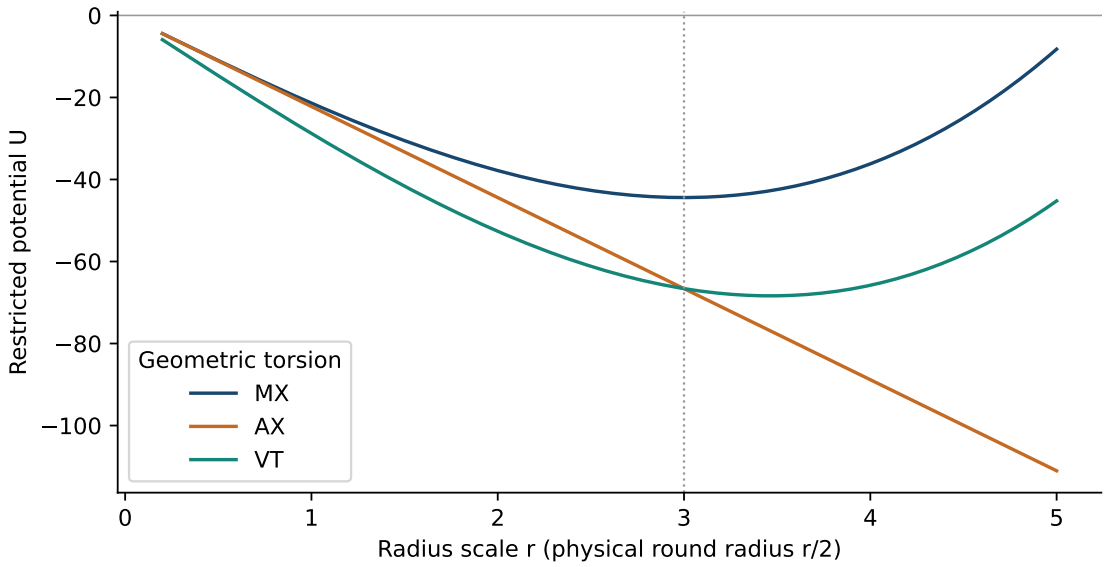


Figure 1: Radius potentials at $L = \kappa = 1$, with each active torsion coefficient set to one. The MX minimum is $r = 3$; the VT minimum is $r = \sqrt{12}$; the AX curve is linear.

The torsion coordinates have positive diagonal static curvature. Their coupling to radius is background dependent:

$$\begin{aligned} U_{\eta\eta} &= \frac{3\pi^2 L r}{2\kappa^2}, \quad U_{VV} = \frac{\pi^2 L r^3}{6\kappa^2}, \qquad U_{\eta V} = 0, \\ U_{r\eta} &= \frac{3\pi^2 L \eta}{2\kappa^2}, \quad U_{rV} = \frac{\pi^2 L r^2 V}{2\kappa^2}. \end{aligned} \tag{S3}$$

Consequently V separates quadratically from (r, η) at an AX point with $V = 0$, whereas it couples to r in VT and MX. The equality $U_{\eta\eta} = U_{VV}$ at $r = 3$ is an equality in these coordinates, not a degeneracy enforced by $SO(3)$. Including L gives further scalar cross terms. Appendix F gives the full gradient and Hessian in one common coordinate system.

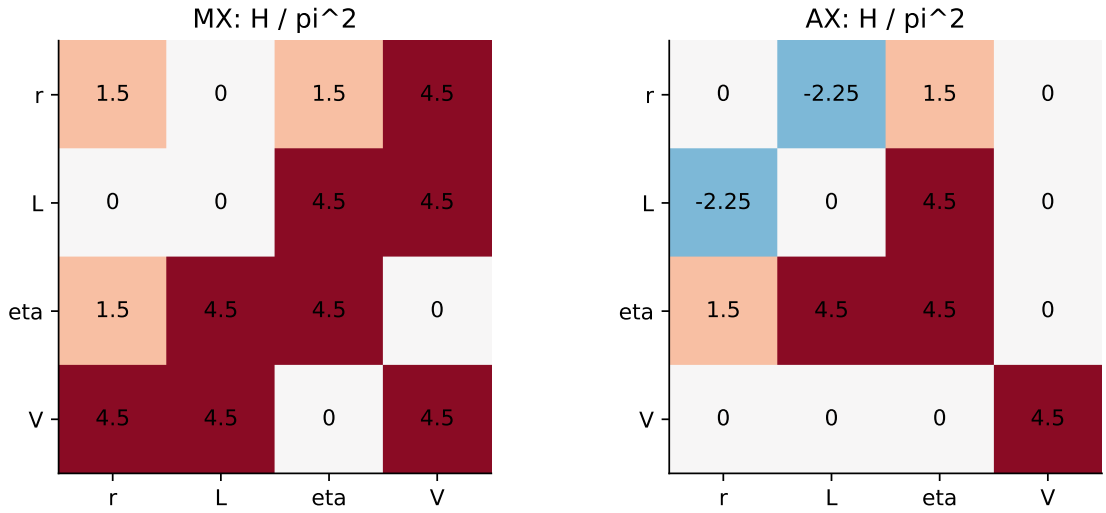


Figure 2: Scalar Hessians in the common coordinates (r, L, η, V), divided by π^2, at the MX point $(3, 1, 1, 1)$ and the AX point $(3, 1, 1, 0)$, with $\kappa = 1$. They display couplings between scalar deformations.

3.2 Exact scalar action along the circle

Allow smooth periodic $r(y), L(y), \eta(y), V(y)$, with $r, L > 0$, on the round zero-twist coframe. Including the connection derivatives and integrating the total derivatives over the circle gives

$$\mathcal{S}_{\text{scalar}} = \int_0^1 dy \left[U_0(r, L, \eta, V) - \frac{3\pi^2}{4\kappa^2} \frac{r(r')^2}{L} \right]. \tag{S4}$$

Appendix A derives this formula directly from Cartan curvature. Radius carries a circle-derivative term. The η' and V' contributions to curvature combine into total derivatives, and no independent L' term remains in the bulk. Here $L(y)$ is the local lapse, while $\int_0^1 L(y)\, dy$ is the circle circumference. Its variation must be specified separately from fixing that circumference. The negative radial derivative coefficient characterizes radial variations of the Euclidean action defined here. The role of lapse and constraints in a canonical formulation is reviewed in [6].

The scalar sector therefore supplies a useful hierarchy: the radius potential at fixed controls, the static response to changing those controls, and the circle-dependent restricted action. For example, the MX radius minimum fixes one derivative to zero, while the remaining scalar gradients need not vanish. This is sufficient for a conditional radius classification and preserves the different roles of size, torsion and circle scale.

In the joint action, scalar changes can couple to shape and twist derivatives through the induced metric (M4). The vanishing of these cross terms in (S4) follows from its round, zero-twist

restriction. At nonzero twist the lapse is $\mathcal{N}$, which differs from the coframe coefficient L; an L' cross term in those coordinates need not represent an independent lapse degree of freedom.

4 Spin-1: twist, mixing and the quartic derivative term

4.1 Static vector geometry

At fixed scalars and shape, the twist term in (M3) is positive definite in ω. At the round point each (ω_i, δ_i) pair has the static Hessian

$$H_{\omega\delta} = \begin{pmatrix} 2\pi^2 L^3/(\kappa^2 r) & 0 \\ 0 & 0 \end{pmatrix}. \tag{V1}$$

The flat mixing direction expresses scalar-action independence under the specified coframe rotation. Curvature tensors need not coincide when torsion components are prescribed anew in the rotated frame. This scalar flatness alone therefore does not identify every mixing variation with a simultaneous gauge transformation of the full fields.

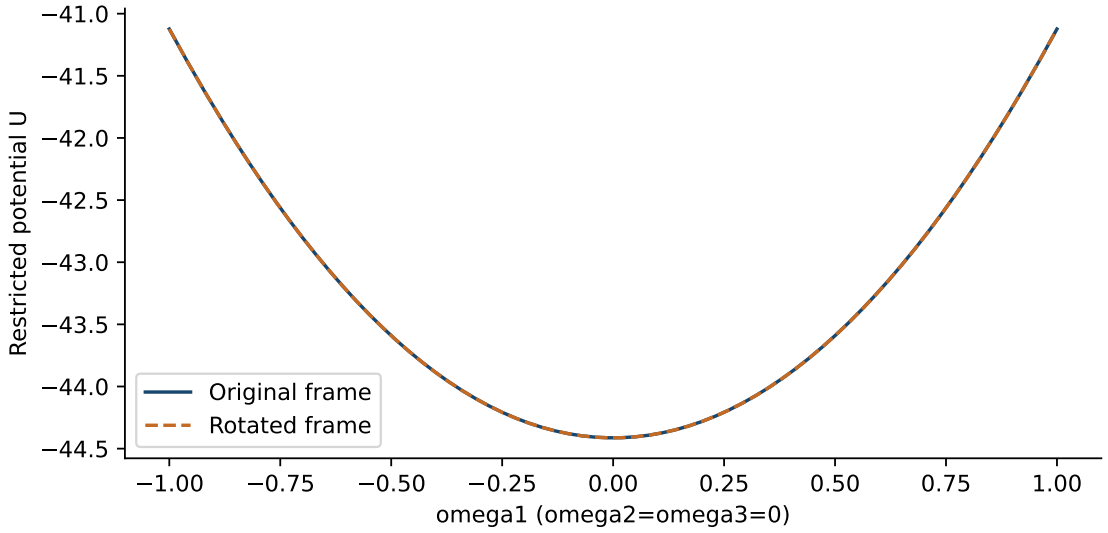


Figure 3: Static twist potential at $r = 3$ and $L = \kappa = \eta = V = 1$. An unrotated and a rotated coframe give the same scalar action.

4.2 Finite-amplitude circle dependence

Set $M = I$ and keep r, L, η, V fixed. For an arbitrary real smooth periodic vector $\omega(y)$, define Q. The complete derivative calculation yields

$$\begin{aligned} Q &= \|\omega\|^2\|\omega'\|^2 - (\omega\cdot\omega')^2 = \|\omega\times\omega'\|^2, \\ R_{\mathrm{LC}} &= \frac{24}{r^2} - \frac{8L^2}{r^4}\|\omega\|^2 - \frac{2}{r^2}\partial_y(\omega\cdot\omega') - \frac{L^2}{2r^4}Q. \end{aligned} \tag{V2}$$

For fixed torsion norms, the remaining EC and complete-NY derivative contributions are total derivatives. A smooth periodic mixing rotation changes only these boundary currents. Subtracting the zero-twist action gives the exact restricted expression

$$\Delta\mathcal{S}_\omega = C_\omega \int_0^1 dy \left(\|\omega\|^2 + \frac{Q}{16}\right), \qquad C_\omega = \frac{\pi^2 L^3}{\kappa^2 r} > 0. \tag{V3}$$

This is valid at finite amplitude, not merely through fourth order in a small-field expansion. The derivative term is quartic in amplitude under $\omega \mapsto t\omega$, but contains only first derivatives squared. With $\omega = \rho n$, $|n| = 1$, it becomes $Q = \rho^4 |n'|^2$: changes of direction carry this cost, whereas changes of magnitude alone do not. The velocity Hessian is

$$G_{ij}(\omega) = \frac{C_\omega}{8}\left(\|\omega\|^2 \delta_{ij} - \omega_i \omega_j\right). \tag{V4}$$

At nonzero twist its radial eigenvalue is zero and its two transverse eigenvalues are $C_\omega |\omega|^2/8$. At zero twist the full quadratic velocity matrix vanishes. This rank statement describes the restricted Euclidean action; the two transverse directions are not a count of physical propagating polarizations.

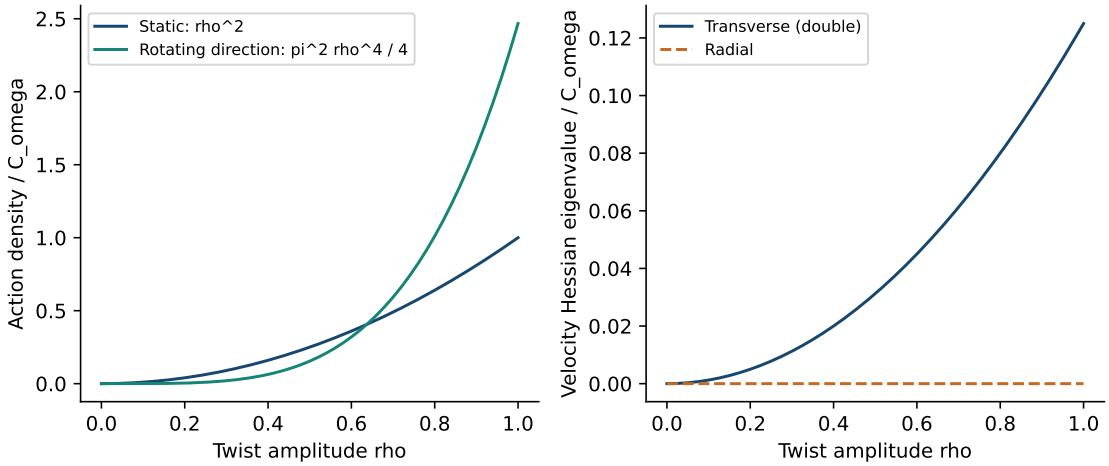


Figure 4: Static and derivative action densities for $\omega = \rho(\cos(2\pi y), \sin(2\pi y), 0)$, and the velocity-Hessian eigenvalues. Quantities are normalized by C_ω. The derivative contribution scales as ρ^4.

4.3 Stationary twist configurations in this family

Let $E_i = \partial\mathcal{L}/\partial\omega_i - \partial_y(\partial\mathcal{L}/\partial\omega_i')$. The radial contraction of the Euler equations is

$$\omega_i E_i = C_\omega \left(2\|\omega\|^2 + \frac{Q}{4}\right). \tag{V5}$$

Since $Q \geq 0$, unrestricted variation of the real twist vector in this fixed-scalar, round-shape family has only $\omega = 0$ as a stationary configuration. It is also the unique minimizer of (V3). Thus the finite-amplitude direction cost and the absence of a quadratic term at the origin fit into one variational structure. Appendix E derives (V2)–(V5); the joint action (M6) determines how this restriction changes when shape also evolves.

4.4 Twist coupling to a rotating shape

For an explicit joint profile take $M = \mathrm{diag}(a, b, (ab)^{-1}) R_{01}(\vartheta(y))$, $\omega = (0, 0, w)$ and fixed r, L, η, V, a, b. With $R_{01}' = R_{01}[\hat{e}_2]_\times \vartheta'$, the last term in (M6) gives

$$\mathcal{L}_{\text{linear}} = -\frac{\pi^2 L r}{2\kappa^2}(a^2 - b^2)^2 \vartheta' w. \tag{V6}$$

At $w = 0$, the twist Euler expression is the coefficient of w in (V6). For a prescribed anisotropic shape with changing orientation it need not vanish. Taking periodic ϑ and $\delta w = \vartheta'$ gives a nonzero first variation, so this term cannot be discarded as a periodic boundary contribution.

This result identifies the coupling that limits the zero-twist stationary statement to its stated family. It does not construct a solution of all joint Euler equations.

5 Spin-2: shape splitting and continuous bifurcation

5.1 The quintet and its axial splitting

Write $h = q_A T_A$, where T_A is the orthonormal symmetric traceless basis of Appendix C. At $\omega = 0$ the scalar controls multiply the shape action by $\mathcal{K}$ and add a shape-independent constant. At the round point its Hessian is $8\mathcal{K}\delta_{AB}$, so all five normalized directions have the same positive curvature. For the axial reference $M_0 = \mathrm{diag}(a, a, a^{-2})$, with $a = (1+\epsilon)^{1/3}$, $\epsilon > -1$, use the paths $M = M_0 e^{q_A T_A}$ one at a time. Their curvatures split into a singlet and two doublets:

$$\begin{aligned} H_0 &= \frac{8\mathcal{K}(4 - a^6)}{3a^8}, \\ H_2 &= \frac{8\mathcal{K}(2a^6 - 1)}{a^2}, \qquad (T_2, T_3), \\ H_1 &= \frac{4\mathcal{K}(a^6 + 1)}{a^8}, \qquad (T_4, T_5). \end{aligned} \tag{T1}$$

At $a = 1$ these agree with the round value. Residual rotations in the 01 plane rotate (T_2, T_3) with twice the angle and (T_4, T_5) with the angle itself. The specified exponential path fixes the meaning of the second derivatives. Off a stationary shape, another path with the same tangent can have a different second derivative, as derived in Appendix D. In particular, the zero of H_0 at $\epsilon = 1$ is a curvature zero along that path.

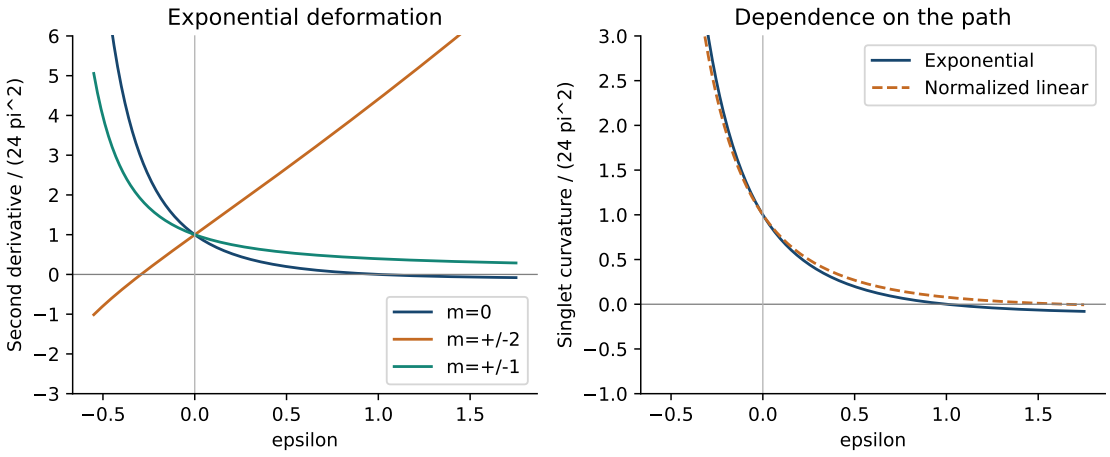


Figure 5: The singlet and two doublet curvatures, normalized by $24\pi^2$ at $r = 3$, $L = \kappa = 1$. The second panel compares two paths with the same tangent at the axial background.

5.2 Exact doublet minimization at fixed axial control

Fix a and minimize over the full doublet, $M = M_0 \exp(q_2 T_2 + q_3 T_3)$. Let $\rho = (q_2^2 + q_3^2)^{1/2}$. The eigenvalues of S are $a^2 e^{\sqrt{2}\rho}, a^2 e^{-\sqrt{2}\rho}, a^{-4}$. Thus $z = 2\cosh(\sqrt{2}\rho) \geq 2$ reduces the problem to a convex quadratic:

$$F(a, z) = a^4(z^2 - 4) - 2a^{-2}z + a^{-8} = a^4(z - a^{-6})^2 - 4a^4. \tag{T2}$$

$$\rho_*(a) = \begin{cases} 0, & a^6 \geq 1/2, \\ \operatorname{arcosh}(1/(2a^6))/\sqrt{2}, & a^6 < 1/2, \end{cases}$$
$$F_{\min}(a) = \begin{cases} a^{-8} - 4a^{-2}, & a^6 \geq 1/2, \\ -4a^4, & a^6 < 1/2. \end{cases} \tag{T3}$$

The critical control is $\epsilon_c = 1/\sqrt{2} - 1$. For $\epsilon < \epsilon_c$ the minimum has three unequal shape eigenvalues; it approaches the axial shape continuously at the boundary. More precisely,

$$\rho_* = 2^{3/4}\sqrt{\epsilon_c - \epsilon} + O((\epsilon_c - \epsilon)^{3/2}),$$
$$[\,\partial_\epsilon^2 F_{\min}\,]_{\text{right}} - [\,\partial_\epsilon^2 F_{\min}\,]_{\text{left}} = 32\,2^{1/3}. \tag{T4}$$

Both $F_{\min}$ and its first derivative are continuous at ϵ_c. The quartic coefficient in the local expansion in ρ is $2\,2^{1/3} > 0$ at criticality. These facts characterize a continuous bifurcation of constrained shape minima, or a continuous shape transition within this finite-dimensional minimization problem.

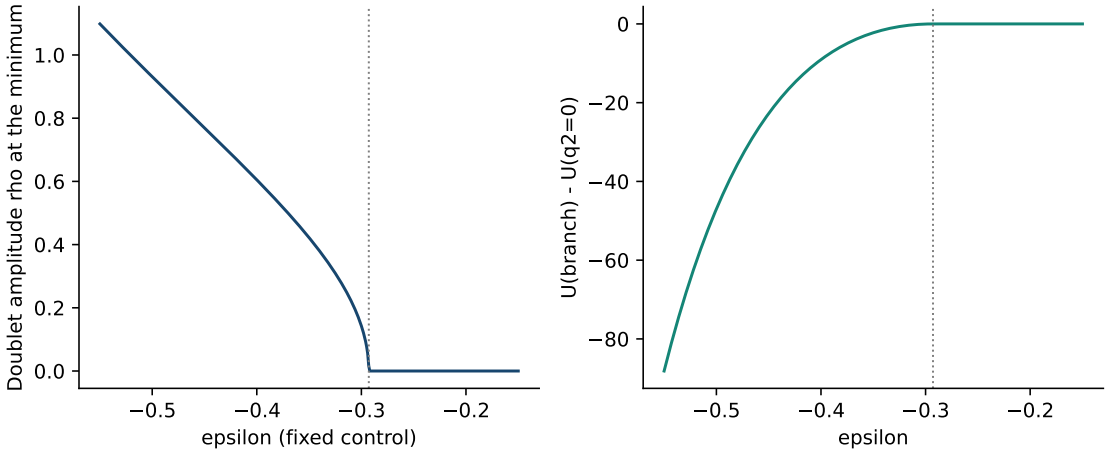


Figure 6: Doublet amplitude and the potential gain at fixed axial control. The dotted line is ϵ_c. For nonzero amplitude the minimizers form a rotation orbit, not two isolated minima.

The two signs of q_2 on the cut $q_3 = 0$ are points on the circle $q_2^2 + q_3^2 = \rho_*^2$. Residual $SO(2)$ rotations connect that circle. When shapes related by rotations and axis relabeling are identified, it represents one triaxial shape class at each fixed control.

5.3 Global minimum of the static shape potential

At fixed scalar controls and zero twist, every positive-definite S of determinant one has an eigenvalue representation of the form used in (T2), after choosing an axis order. Minimizing (T3) also over $a > 0$ therefore covers all shapes, modulo rotations. For $a^6 < 1/2$, $F_{\min} > -2^{4/3} > -3$. On the complementary branch,

$$F_{\min} + 3 = \frac{(a^2-1)^2(3a^4 + 2a^2 + 1)}{a^8} \geq 0. \tag{T5}$$

Equality requires $a = 1, z = 2$, hence $S = I$. On this zero-twist slice, the round shape is the unique global minimizing shape of the static potential, modulo rotations. Since the twist term in (M3) is nonnegative, minimizing over twist as well gives $\omega = 0$. At a prescribed nonzero twist, the shape dependence of the last term in (M3) generally shifts the shape minimum. The fixed-control bifurcation and the full-shape minimum above are complementary results of the

static zero-twist potential. At fixed torsion and circle circumference, combining this bound with Section 3 also selects the round radius minimum when $V \neq 0$ and $\eta^2 < 4$.

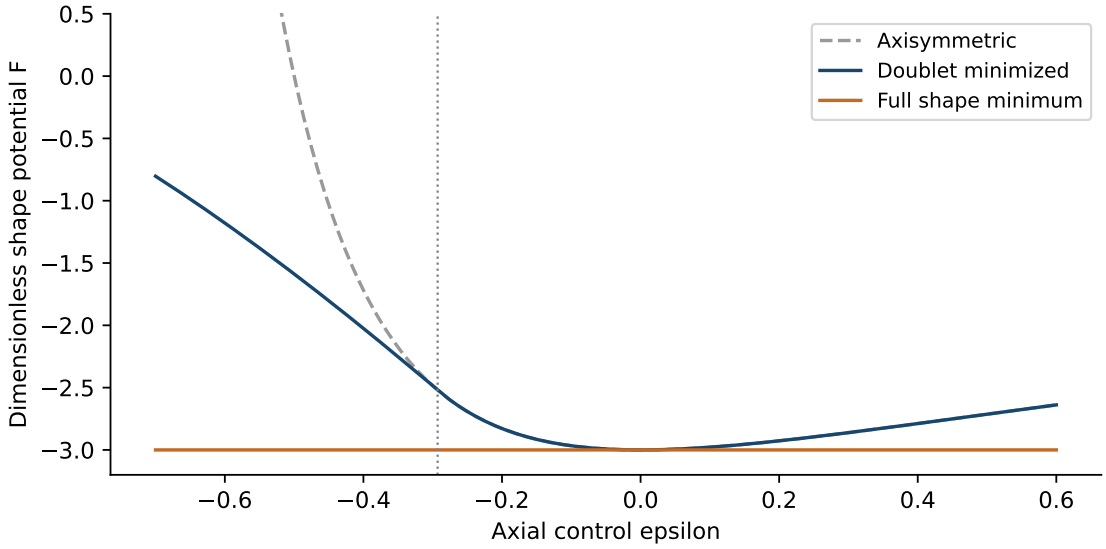


Figure 7: The axial shape potential, its doublet-minimized value and the full-shape lower bound $F = -3$. Releasing the axial control selects $\epsilon = 0$, $\rho = 0$.

5.4 Circle-dependent shape

Set $\omega = 0$ and write $B = M^{\mathsf{T}}M$, with $\det B = 1$. Then $Y = 0$ and $\mathrm{tr}(B^{-1}B') = 0$. The derivative part of (M6) becomes

$$\mathcal{L}_{\partial}\big|_{\omega=0} = -\frac{3\pi^2 r(r')^2}{4\kappa^2 L} + \frac{\pi^2 r^3}{32\kappa^2 L}\mathrm{tr}\Big[(B^{-1}B')^2\Big]. \tag{T6}$$

The shape term is nonnegative: $B^{-1}B'$ is similar to the symmetric matrix $B^{-1/2}B'B^{-1/2}$. Near a constant round reference, $M = e^h$ gives the quadratic coefficient $\pi^2 r_0^3 \mathrm{tr}[(h')^2]/(8\kappa^2 L_0)$. The scalar–shape derivative cross term vanishes exactly at zero twist. Thus the static bifurcation and a positive shape derivative cost can be described in the same sector. Their extension to nonzero twist requires the joint terms of Section 6.

6 Interactions, coupled profiles and curvature identities

6.1 Interactions between irreducible sectors

For $M = e^h$, define $I_2 = \mathrm{tr}(h^2)$ and $I_3 = \mathrm{tr}(h^3)$. Cayley–Hamilton gives $\mathrm{tr}(h^4) = I_2^2/2$. Expanding the exact potential yields

$$\begin{aligned} F(e^{2h}) &= -3 + 4I_2 + \frac{40}{3}I_3 + \frac{14}{3}I_2^2 + O(h^5), \\ \|e^h\omega\|^2 &= \omega^2 + 2\omega^{\mathsf{T}}h\omega + 2\omega^{\mathsf{T}}h^2\omega + O(h^3\omega^2). \end{aligned} \tag{I1}$$

The first line is multiplied by $\mathcal{K}$, and the second by $\mathcal{K}L^2/r^2$. It supplies the tensor self-interactions and tensor–twist couplings in the same normalization as the separate sectors. A cubic tensor invariant is generally present: the even radial doublet potential is a property of its specified slice, not an $h \mapsto -h$ symmetry of the full model. Representation theory identifies allowed invariants, while the explicit action determines their coefficients. Appendix G gives the expansion and its relation to the quadratic blocks.

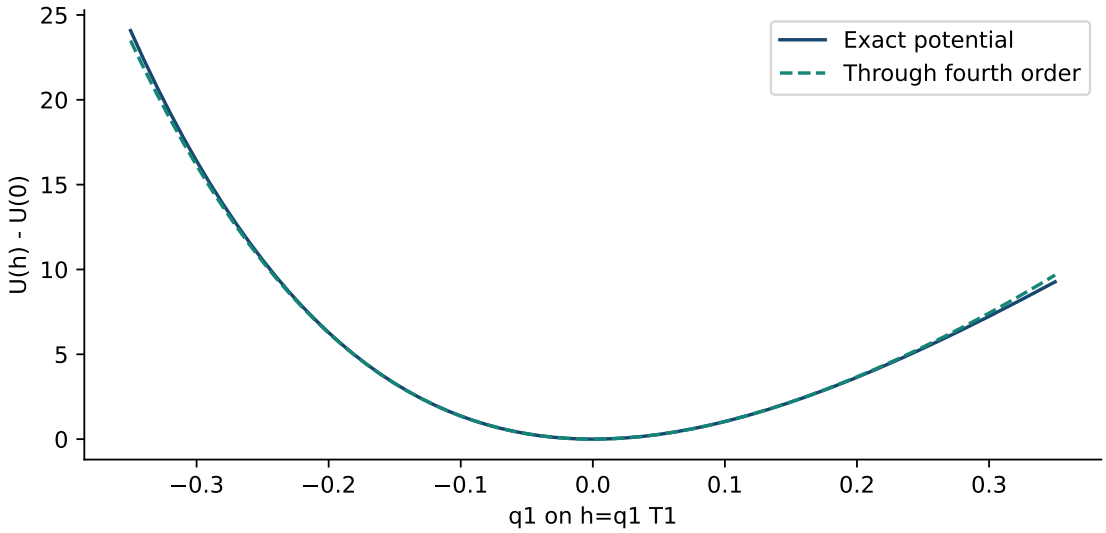


Figure 8: Exact shape-potential difference along $h = q_1 T_1$ and its invariant expansion through fourth order, at fixed $r = 3$, $L = \kappa = \eta = V = 1$.

6.2 Derivative cross terms

The joint action preserves quadratic separation at a constant round, zero-twist reference. Counting every perturbation and its derivative as first order, representative cross-sector cubic terms are

$$\begin{aligned}\mathcal{L}^{(3)}_{\text{cross}} = & -\frac{\pi^2 L_0}{2\kappa^2}\delta r'\,\omega^{\mathsf{T}}\omega' + \frac{\pi^2 L_0 r_0}{4\kappa^2}\omega'^{\mathsf{T}} h'\omega \\ & -\frac{\pi^2 L_0 r_0}{\kappa^2}\operatorname{tr}\left[h'\left(h[\omega]_\times - [\omega]_\times h\right)\right].\end{aligned} \tag{X1}$$

These terms include two velocity cross terms and the leading shift coupling. Expanding scalar-dependent coefficients produces further vertices, all contained in (M6). For finite-amplitude examples set $M = \operatorname{diag}(e^s, e^{-s}, 1)$ and $\omega = (w, 0, 0)$. With $\mathsf{G}_{ab} = \partial^2\mathcal{L}_\partial/\partial x'_a \partial x'_b$, direct differentiation gives

$$\begin{aligned}\mathsf{G}_{rw} &= -\frac{\pi^2 L w}{2\kappa^2 e^{2s}}, \quad \mathsf{G}_{sw} = \frac{\pi^2 L r w}{4\kappa^2 e^{2s}}, \quad \mathsf{G}_{rs} = \frac{\pi^2 L w^2}{4\kappa^2 e^{2s}}, \\ \mathsf{G}_{rL} &= -\frac{\pi^2 w^2}{2\kappa^2 e^{2s}}, \quad \mathsf{G}_{sL} = \frac{\pi^2 r w^2}{4\kappa^2 e^{2s}}.\end{aligned} \tag{X2}$$

The cross terms are nonzero at nonzero twist. A velocity Hessian alone does not include the velocity-linear term (V6), which must also be retained in the Euler equations. Appendix J gives the general velocity rank and the distinction from a physical degree-of-freedom count.

6.3 A coupled periodic family and absence of a lower bound

Keep r, L, η, V fixed in the diagonal family above. The derivative density has the exact completed-square form

$$\begin{aligned}\mathcal{L}_\partial &= \frac{\pi^2 r^3}{4\kappa^2 L}(s')^2 + \frac{\pi^2 L r}{4\kappa^2 e^{2s}} s' w w' \\ &= \frac{\pi^2 r^3}{4\kappa^2 L}\left(s' + \frac{L^2 w w'}{2r^2 e^{2s}}\right)^2 - \frac{\pi^2 L^3 w^2 (w')^2}{16\kappa^2 r e^{4s}}.\end{aligned} \tag{X3}$$

At fixed shape this one-axis twist has zero derivative cost. Allowing the shape to follow the twist produces the negative second term. Choose the smooth periodic, nondegenerate family

$$w_n(y) = A\cos(2\pi n y), \quad e^{2s_n} = 1 - \frac{L^2 w_n^2}{2r^2}, \quad 0 < A < \frac{\sqrt{2}r}{L}, \quad n = 1, 2, \ldots,$$
$$\mathcal{S}_n = \overline{U}_A - n^2 C_A, \qquad C_A = \frac{\pi^2 L^3}{16\kappa^2 r}\int_0^1 \frac{w_1^2 (w_1')^2}{[1 - L^2 w_1^2/(2r^2)]^2}\, dy > 0. \tag{X4}$$

The relation for s_n makes the square in (X3) vanish. Since each profile repeats the same path n times, the static average $\overline{U}_A = \int_0^1 U(r, L, \eta, V, M_n, \omega_n)\, dy$ is independent of n, while the derivative contribution scales as $-n^2$. Hence $\mathcal{S}_n \to -\infty$. The simultaneous-profile Euclidean action is unbounded below even at fixed scalar controls, in the model with no higher-derivative term or frequency cutoff. This unboundedness concerns the Euclidean action on the specified smooth periodic profiles before imposing the constraints.

The negative contribution has a geometric origin. With $\widetilde{X} = X - \frac{1}{3}(\mathrm{tr}X)I$, the velocity quadratic form in (M6) is $\mathrm{tr}(\widetilde{X}^2) - \frac{2}{3}(\mathrm{tr}X)^2$, where $\mathrm{tr}X = (\ln\det\gamma)'$ measures the change of the induced spatial volume. Fixing r and L does not fix this volume: along (X4), $\det\gamma = r^6(1 + L^2 w_n^2 e^{-2s_n}/r^2)$ varies with the twist, and the trace term dominates wherever $w_n w_n' \neq 0$. The negative radial term in (S4) has the same trace origin. The sequence thus realizes, at fixed scalar controls and through the shape–twist coupling, the conformal-mode indefiniteness of the Euclidean gravitational action [7].

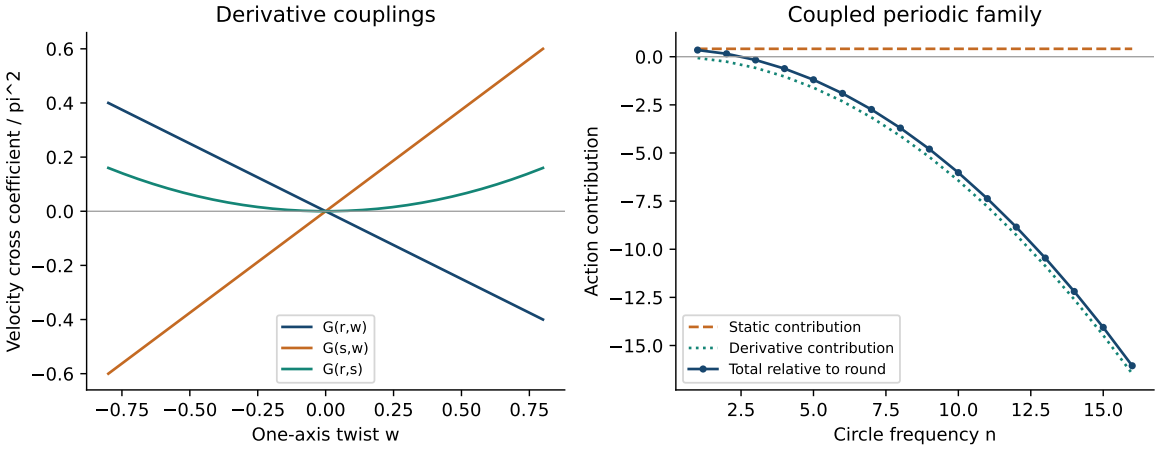


Figure 9: Left: velocity cross coefficients (X2) at $r = 3$, $L = \kappa = 1$, $s = 0$. Right: action contributions relative to the round static configuration for (X4), with $A = 0.5$ and $\eta = V = 1$. The static contribution is independent of frequency, while the coupled derivative contribution decreases as $-n^2$.

For these parameters the round static action is -44.4132198049. The joint action is -44.0662455826 at $n = 1$, -45.0308924809 at $n = 4$ and -48.1177625553 at $n = 8$. Thus the static global shape bound of Section 5 and the absence of a lower bound for the circle functional coexist: they minimize different objects over different families.

6.4 Complete NY and form-index Pontryagin

Use the full EC curvature, with no pair-exchange symmetry assumed. The densities are

$$
\begin{aligned}
N &= \frac{1}{4}\epsilon^{abcd}T^{e}{}_{ab}T_{ecd} - \frac{1}{2}\epsilon^{abcd}R_{abcd},\\
P &= \frac{1}{4}\epsilon^{cdef}R_{abcd}R_{abef}, \qquad P\,\mathrm{vol}_4 = \sum_{a,b} R_{ab}\wedge R_{ab},\\
B &= \sum_{c<d}\left(R_{01cd}R_{23cd} - R_{02cd}R_{13cd} + R_{03cd}R_{12cd}\right).
\end{aligned}
\tag{I2}
$$

For any constant connection on an oriented four-dimensional unimodular Lie algebra, every left-invariant three-form is closed. Write such a form as $\iota_X\mathrm{vol}$; its exterior derivative is $\mathcal{L}_X\mathrm{vol}$, which vanishes by unimodularity. Both the NY three-form and the connection Chern–Simons three-form, whose exterior derivatives give the two invariants on Riemann–Cartan spaces [5], are left invariant. Therefore

$$
\begin{aligned}
N\,\mathrm{vol}_4 &= d(e^a\wedge T_a) = 0,\\
P\,\mathrm{vol}_4 &= -d\,\mathrm{tr}\left(\Gamma\wedge d\Gamma + \frac{2}{3}\Gamma\wedge\Gamma\wedge\Gamma\right) = 0.
\end{aligned}
\tag{I3}
$$

Here Γ in the second line denotes the matrix-valued connection one-form. The theorem includes AX, VT and MX and arbitrary constant coframe deformations of $S^3\times S^1$. It is a local algebraic statement; compactness supplies the integration domain. The internal-dual quantity B is a different contraction and may be nonzero. Appendix B gives its independent normalization and a nonconstant example for which $P\neq 0$.

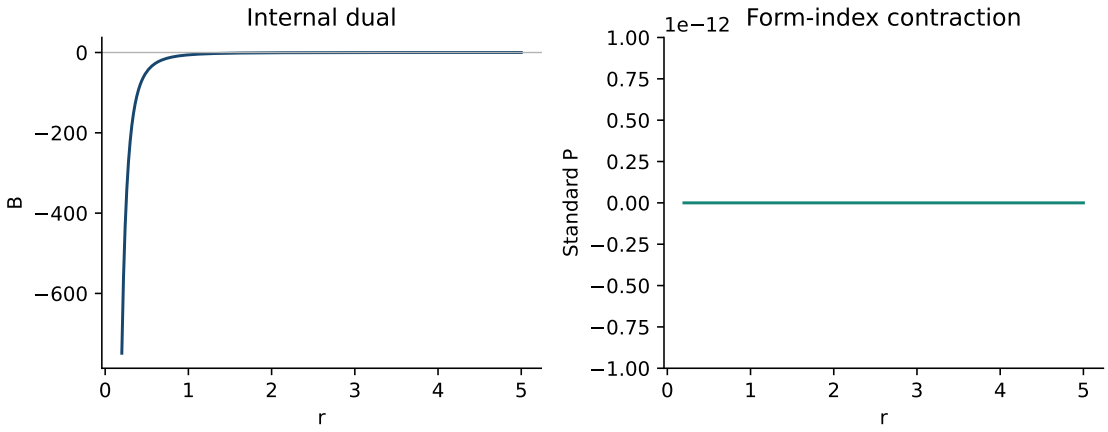


Figure 10: Internal-dual B and standard form-index P on the round MX family with $\eta = V = 1$. The standard density vanishes analytically; its plotted values show floating-point residuals.

7 Geometric interpretation and relation to Papers 01 and 02

The spin sectors provide one organization for two variational objects. The static potential determines size–torsion couplings, twist stiffness and shape minima. The circle action determines the derivative costs and cross terms when those coordinates vary. Spin-0 carries the radial term, spin-2 carries the tracefree shape term, and spin-1 acquires its finite-amplitude direction cost on the fixed-shape restriction. Their common metric dependence produces the nonlinear couplings.

At fixed scalar controls and zero twist, holding the axial control fixed in the static potential exposes a continuous doublet bifurcation; releasing that control selects the round shape as the unique global minimum, modulo rotations. The positive factor $\mathcal{K}$ and additive torsion terms leave this dimensionless classification unchanged at fixed scalars. Static minimization over twist also selects $\omega = 0$. On the common spherical family this recovers Paper01's FULL radius potential

and Paper02's static axisymmetric bound at $\alpha = 0$, while extending the shape analysis to all five traceless components. Paper02's joint minimization varies constant radius and shape parameters; the joint circle problem here instead varies periodic profiles.

For the circle functional, (X4) gives a different result: the same static bound coexists with a sequence whose action tends to minus infinity. Nonnegative derivative terms on the zero-twist, fixed-radius shape restriction and the fixed-scalar, round-shape twist restriction do not imply positivity after coupling them. Likewise, the zero-twist stationary result applies to the family of Section 4.2; the shape-rotation source (V6) explains why a prescribed general shape history need not preserve it. This comparison identifies the contribution of each sector without identifying its coordinates with independent particles.

All extrema retain their specified controls. At the MX reference point the radial derivative vanishes but other scalar gradients do not, so its radius minimum is a restricted model result. The non-bounded circle functional is also an off-shell Euclidean result, established before imposing lapse and momentum constraints or choosing a Lorentzian continuation. No vacuum or physical response observable is inferred. Paper02 includes an irreducible EC-Weyl correction; its coefficient is zero here. Its $\alpha > 0$ unboundedness arises from static small-radius limits. Our (X4) is instead a large-frequency sequence at fixed positive radius and $\alpha = 0$. The equality of the two Weyl projections on Paper02's prescribed static family is not used as an identity for circle-dependent fields. Whether additional terms or a specified frequency range change the circle problem is a different model question.

8 Conclusions

Einstein–Cartan theory with constant complete NY on $S^3 \times S^1$ admits a common static potential and an exact joint circle action organized by spin-0, spin-1 and spin-2 geometric deformations. The scalar sector contains radius–torsion couplings and a conditional radius minimum. The vector sector has a positive static twist cost and a positive amplitude-quartic direction term on its fixed-scalar, round-shape restriction, whose unique stationary twist is zero. The tensor sector has a nonnegative derivative term on the zero-twist, fixed-radius circle restriction. Its constrained static minima undergo a continuous bifurcation with a square-root amplitude law and a second-derivative jump.

At fixed scalar controls, minimizing the static potential jointly over all constant volume-preserving shapes and twist selects the round geometry with zero twist. The joint circle action has a different global property: derivative cross terms and shift couplings connect the sectors, and a smooth periodic shape–twist family makes the action unbounded below even at fixed scalar controls. These are compatible results for different variational objects. The general action recovers the separate scalar and vector formulas, while exhibiting the nonlinear terms that control their joint variation. Constant-connection curvature identities and periodic NY boundary cancellation complete the geometric description.

The calculation code, saved data and figure reproduction instructions are described in Appendix H.

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The AI tools did not determine the scientific claims of this work; numerical results and figures were produced by deterministic calculations of the stated formulas, rather than by generative image synthesis. The author takes full responsibility for the content and for any remaining errors.

For this revision, Codex (OpenAI; accessed 2026-11) also assisted with computational and plotting code, bilingual presentation, and symbolic and numerical checks.

This work was developed within the informal collaborative project **DPPU** (**D**onut-like topology, **P**lanck-scale compactness, **P**recession dynamics, **U**niverse).

A Cartan connection and the scalar circle action

Use $de^a = \frac{1}{2}C^a{}_{bc}e^b \wedge e^c$ and $\Gamma^a{}_b = \Gamma^a{}_{bc}e^c$. Metric compatibility and the specified geometric torsion give

$$\begin{aligned} \Gamma^{\mathrm{LC}}_{abc} &= \tfrac{1}{2}(C_{abc} + C_{cba} - C_{bac}), \\ K_{abc} &= -\tfrac{1}{2}(T_{abc} + T_{bca} - T_{cab}), \qquad \Gamma^{\mathrm{EC}} = \Gamma^{\mathrm{LC}} + K, \\ R^a{}_{bcd} &= e_c(\Gamma^a{}_{bd}) - e_d(\Gamma^a{}_{bc}) + \Gamma^a{}_{ec}\Gamma^e{}_{bd} - \Gamma^a{}_{ed}\Gamma^e{}_{bc} + \Gamma^a{}_{be}C^e{}_{cd}. \end{aligned} \tag{A1}$$

Substitution verifies $C^a{}_{bc} + \Gamma^a{}_{cb} - \Gamma^a{}_{bc} = T^a{}_{bc}$. For $e = A(y)\bar{e}$, with $\bar{e} = (\sigma^0, \sigma^1, \sigma^2, dy)$, the full exterior derivative is

$$\begin{aligned} C^a{}_{bc} &= A^a{}_d(A^{-1})^e{}_b(A^{-1})^f{}_c\bar{C}^d{}_{ef} \\ &\quad + (\partial_y A^a{}_f)\left[(A^{-1})^3{}_b(A^{-1})^f{}_c - (A^{-1})^3{}_c(A^{-1})^f{}_b\right], \\ e_c(\Gamma^a{}_{bd}) &= (A^{-1})^3{}_c\partial_y\Gamma^a{}_{bd}. \end{aligned} \tag{A2}$$

For constant $A = O\left(\begin{smallmatrix} rM & 0 \\ L\omega^{\mathsf{T}} & L \end{smallmatrix}\right)$, these equations reduce to the constant-coframe algebra. With $S = MM^{\mathsf{T}}$ their scalar contraction is

$$R = \frac{8}{r^2}\{[\mathrm{tr}S]^2 - 2\mathrm{tr}(S^2)\} - \frac{8L^2}{r^4}\|M\omega\|^2 - \frac{6\eta^2}{r^2} - \frac{2V^2}{3}. \tag{A3}$$

For diagonal $M = \mathrm{diag}(f_0, f_1, f_2)$, $\prod_i f_i = 1$, the spatial structure constants are $4f_i^2\epsilon_{ijk}/r$. Their substitution gives the shape term in (A3); the matrix transformation (A2) supplies the general coframe and twist. Multiplying by $-\pi^2 Lr^3/(8\kappa^2)$ proves (M3). For the scalar circle family, direct use of (A1)–(A2) instead gives

$$\begin{aligned} R_{\mathrm{EC}} &= \frac{24 - 6\eta^2}{r^2} - \frac{2V^2}{3} - \frac{6r''}{L^2 r} - \frac{6(r')^2}{L^2 r^2} + \frac{6r'L'}{L^3 r} + \frac{2}{L}\left(V' + \frac{3r'V}{r}\right), \\ N &= -\frac{6}{Lr^3}\partial_y(\eta r^2). \end{aligned} \tag{A4}$$

To isolate the bulk, multiply by the volume density. With $R_0 = (24 - 6\eta^2)/r^2 - 2V^2/3$,

$$
\begin{aligned}
Lr^3 R_{\mathrm{EC}} &= Lr^3 R_0 + \frac{6r(r')^2}{L} + \partial_y\left(-\frac{6r^2 r'}{L} + 2r^3 V\right), \\
Lr^3 N &= -6\partial_y(\eta r^2).
\end{aligned}
\tag{A5}
$$

The boundary terms integrate to zero for smooth periodic profiles. This proves (S4), including its sign and normalization. The torsion derivative terms have not been omitted; their contribution is contained in the displayed total derivatives.

B Curvature contractions and a nonconstant example

For a two-form F, $F \wedge F = 2(F_{01}F_{23} - F_{02}F_{13} + F_{03}F_{12})\mathrm{vol}_4$. The standard form-index density therefore also has the row-Pfaffian expression

$$
P = 2\sum_{a,b}\left(R_{ab01}R_{ab23} - R_{ab02}R_{ab13} + R_{ab03}R_{ab12}\right). \tag{B1}
$$

The internal pair is ordered in this sum. The raw full-component Hodge inner product equals $2P$, and $p_1 = P\,\mathrm{vol}_4/(8\pi^2)$ in this convention. Internal-pair and form-pair dualization are different operations for a generic torsionful curvature. The quantity B in (I2) uses the former contraction with the normalization $\frac{1}{4}\epsilon^{abef}R_{abcd}R_{efcd} = 4B$. The left-hand side is the internal contraction quoted in Paper02; its value $-16/27$ at geometric $\eta = V = -1, r = 3$ corresponds to $B = -4/27$ here. The standard form-index P agrees directly between the two papers.

As a nonconstant example, take a Euclidean coordinate coframe and $\Gamma_{ijk} = -f(y)\epsilon_{ijk}$ with $y = x^3$, all other components zero. Then $T_{ijk} = 2f\epsilon_{ijk}$, $R_{0101} = -f^2$ and $R_{0123} = f'$. The three independent spatial internal pairs give

$$
B = 0, \qquad P = -12f^2 f'. \tag{B2}
$$

This illustrates the constant-connection hypothesis of (I3) and distinguishes B from P without assigning a physical response to either. Reversing the sign of the specified connection reverses the sign of the odd expression, consistently with (A1).

C Normalized tensor basis and rotation orbits

Let E_{ij} denote matrix units. The basis is

$$
\begin{aligned}
T_1 &= \frac{\mathrm{diag}(1,1,-2)}{\sqrt{6}}, \quad T_2 = \frac{\mathrm{diag}(1,-1,0)}{\sqrt{2}}, \\
T_3 &= \frac{E_{01}+E_{10}}{\sqrt{2}}, \qquad T_4 = \frac{E_{02}+E_{20}}{\sqrt{2}}, \qquad T_5 = \frac{E_{12}+E_{21}}{\sqrt{2}}.
\end{aligned}
\tag{C1}
$$

They obey $\mathrm{tr}(T_A T_B) = \delta_{AB}$. Conjugation by a rotation in the 01 plane leaves T_1 fixed, rotates (T_2, T_3) by twice its angle and (T_4, T_5) by its angle. Thus the real spin-2 representation decomposes as $1+2+2$ on an axial background. For the doublet, the 01 block of h has eigenvalues $\pm\rho/\sqrt{2}$. It commutes with M_0 in that block, proving the eigenvalue representation preceding (T2).

The diagonal coordinate s used by `shape_matrix` obeys $M = \mathrm{diag}(a(1+s), a/(1+s), a^{-2})$. Its exact relation to the normalized T_2 coordinate is $1 + s = e^{q_2/\sqrt{2}}$. On the cut $q_3 = 0$, a 90-degree rotation exchanges the two axes and maps q_2 to $-q_2$. Thus the two signed branches are part of one residual rotation orbit.

D Shape derivatives and critical behavior

Put $D = M_0^2$ and use $M = M_0 e^{qT}$. Cyclicity gives $A(q) = \mathrm{tr}(De^{2qT})$ and $B(q) = \mathrm{tr}(De^{2qT}De^{2qT})$. At $q = 0$,

$$\begin{aligned} &A_0 = \mathrm{tr}D, && A_0' = 2\mathrm{tr}(DT), && A_0'' = 4\mathrm{tr}(DT^2),\\ &B_0' = 4\mathrm{tr}(D^2T), && B_0'' = 8\mathrm{tr}(D^2T^2) + 8\mathrm{tr}(DTDT), && \\ &F_0'' = 2B_0'' - 2(A_0')^2 - 2A_0A_0''. && && \end{aligned} \tag{D1}$$

Substituting (C1) proves (T1). The normalized linear path $M = M_0(I + qT)/\det(I + qT)^{1/3}$ instead has $\log(M_0^{-1}M) = qT - q^2(T^2 - I/3)/2 + O(q^3)$. Its curvatures are

$$\begin{aligned} \widetilde{H}_0 &= \frac{4\mathcal{K}(7 - a^6)}{3a^8},\\ \widetilde{H}_2 &= \frac{4\mathcal{K}(4a^6 - 1)(3a^6 - 1)}{3a^8},\\ \widetilde{H}_1 &= \frac{2\mathcal{K}(7a^6 + 5)}{3a^8}. \end{aligned} \tag{D2}$$

Both paths have the same first tangent but different acceleration. The chain rule $d^2U/dq^2 = U_{,\mu\nu}x'^{\mu}x'^{\nu} + U_{,\mu}x''^{\mu}$ explains their difference. The acceleration term vanishes at the stationary round shape.

For the doublet let $u = a^6 = (1+\epsilon)^2$. The convex form (T2), with $z \geq 2$, places the minimum at $z = 2$ for $u \geq 1/2$ and at $z = u^{-1}$ otherwise. Expanding $z = 2\cosh(\sqrt{2}\rho)$ gives

$$F(a,\rho) = F(a,0) + \frac{4(2a^6 - 1)}{a^2}\rho^2 + \frac{2(8a^6 - 1)}{3a^2}\rho^4 + O(\rho^6). \tag{D3}$$

At criticality the quadratic coefficient is zero and the quartic coefficient is $2\,2^{1/3}$. Since $1/[2(1+\epsilon)^2] = 1 + 2\sqrt{2}(\epsilon_c - \epsilon) + O((\epsilon_c - \epsilon)^2)$, the exact stationary condition proves the square-root law. Differentiating the two branches of (T3) proves their first-derivative matching and the jump in (T4). Finally, the potential gain below criticality is $F_{\min} - F(a,0) = -(1 - 2a^6)^2/a^8$.

E Derivation of the nonlinear twist action

For the fixed-scalar, round-shape family of Section 4.2 in the unrotated coframe the induced spatial metric, lapse and shift in the invariant basis are

$$\begin{aligned} \gamma &= r^2 I + L^2\omega\omega^{\mathsf{T}}, \qquad \mathcal{N} = \frac{Lr}{\sqrt{r^2 + L^2\omega^2}},\\ \beta^i &= \frac{L^2\omega_i}{r^2 + L^2\omega^2}, \qquad \gamma^{-1} = \frac{1}{r^2}\left(I - \frac{L^2\omega\omega^{\mathsf{T}}}{r^2 + L^2\omega^2}\right). \end{aligned} \tag{E1}$$

At each circle coordinate, the spatial shift is along the symmetry axis of γ and its spatial Lie derivative of γ vanishes. Thus $K = \gamma'/(2\mathcal{N})$ up to an irrelevant overall sign convention for the normal. Direct matrix traces give

$$\begin{aligned} \mathrm{tr}\Big[(\gamma^{-1}K)^2\Big] - \Big[\mathrm{tr}(\gamma^{-1}K)\Big]^2 &= \frac{L^2}{2r^4}Q,\\ {}^{(3)}R &= \frac{24}{r^2} - \frac{8L^2}{r^4}\omega^2. \end{aligned} \tag{E2}$$

The Euclidean curvature decomposition adds the negative of the first line and the divergence $-2\partial_y(\omega\cdot\omega')/r^2$ to ${}^{(3)}R$, giving (V2). Equations (A1)–(A2) give the same result without a metric decomposition. This comparison fixes both the sign and the normalization of the quartic term.

For an arbitrary smooth mixing rotation, let E_3^y denote the circle component of the final frame vector dual to e'^3. With fixed r, L, η, V the torsion decomposition gives

$$\begin{aligned} R_{\mathrm{EC}} &= R_{\mathrm{LC}} - \frac{6\eta^2}{r^2} - \frac{2V^2}{3} + 2V\partial_y E_3^y, \\ N &= -\frac{6\eta}{r}\partial_y E_3^y. \end{aligned} \tag{E3}$$

The volume density is constant and the spatial invariant divergence is zero. The last terms therefore integrate to zero for periodic profiles. For a single 03 rotation and $\omega = (w, 0, 0)$, $E_3^y = \cos\chi/L + w\sin\chi/r$, which exhibits the boundary current explicitly. This proves scalar-action independence from mixing in the circle family.

For completeness, with $p = \omega'$, $s = \omega\cdot p$, the derivative momentum and Euler expression are

$$\begin{aligned} \frac{\partial\mathcal{L}}{\partial p} &= \frac{C_\omega}{8}(\omega^2 p - s\omega), \\ E &= 2C_\omega\omega + \frac{C_\omega}{8}\left[(2p^2 + \omega\cdot\omega'')\omega - 2sp - \omega^2\omega''\right]. \end{aligned} \tag{E4}$$

Contracting with ω cancels the second derivatives and gives (V5). This is a pointwise proof, valid also where the polar coordinate ρ, n is singular. It requires no division by the amplitude and establishes the zero-twist stationary result for all real smooth twist profiles in this fixed-scalar, round-shape family.

F Scalar derivatives and quadratic representation blocks

For $x = (r, L, \eta, V)$, differentiating (S1) yields

$$\nabla U_0 = \frac{\pi^2}{\kappa^2}\begin{pmatrix} L(3V^2r^2 + 9\eta^2 - 36)/12 \\ r(V^2r^2 + 9\eta^2 - 36)/12 \\ 3Lr\eta/2 \\ Lr^3V/6 \end{pmatrix}. \tag{F1}$$

$$H = \frac{\pi^2}{\kappa^2}\begin{pmatrix} LV^2r/2 & (3V^2r^2 + 9\eta^2 - 36)/12 & 3L\eta/2 & LVr^2/2 \\ (3V^2r^2 + 9\eta^2 - 36)/12 & 0 & 3r\eta/2 & Vr^3/6 \\ 3L\eta/2 & 3r\eta/2 & 3Lr/2 & 0 \\ LVr^2/2 & Vr^3/6 & 0 & Lr^3/6 \end{pmatrix}. \tag{F2}$$

At MX $(r, L, \eta, V, \kappa) = (3, 1, 1, 1, 1)$ the gradient is $\pi^2(0, -9/2, 9/2, 9/2)$; at AX $(3, 1, 1, 0, 1)$ it is $\pi^2(-9/4, -27/4, 9/2, 0)$. The nonzero gradients must accompany an interpretation of these matrices. In particular the AX (r, η) determinant is negative, but the point is not a stationary saddle of the full scalar potential. Ordinary eigenvalues of a Hessian at such a point depend on the chosen coordinates and are not particle masses.

At $M = I, \omega = 0$, the first shape derivative of (M3) is zero for every scalar coordinate. Differentiating it again with respect to any scalar proves all scalar–shape cross terms vanish. The twist contribution is even and quadratic in ω, so its first derivative is zero there for every scalar and shape. Mixing does not enter the static scalar action. This directly gives the irreducible quadratic organization

$$(r, L, \eta, V) \ \oplus\ (\omega_i, \delta_i) \ \oplus\ (q_A), \tag{F3}$$

with scalar mixing retained inside the first block, (V1) in the second, and $8\mathcal{K}I_5$ in the third. This is an organization of the static quadratic form. It does not assume an independent kinetic term for every entry. In (S4) the bulk scalar velocity Hessian is $\mathrm{diag}(-3\pi^2 r/(2\kappa^2 L), 0, 0, 0)$, consistent with the different roles of these coordinates.

G Invariant expansion

For a traceless symmetric 3 by 3 matrix, Cayley–Hamilton gives $h^3 - I_2 h/2 - I_3 I/3 = 0$. Multiplication by h and tracing yields $\mathrm{tr}(h^4) = I_2^2/2$. Expanding the exponentials,

$$\begin{aligned} \mathrm{tr}(e^{2h}) &= 3 + 2I_2 + \frac{4}{3}I_3 + \frac{1}{3}I_2^2 + O(h^5), \\ \mathrm{tr}(e^{4h}) &= 3 + 8I_2 + \frac{32}{3}I_3 + \frac{16}{3}I_2^2 + O(h^5). \end{aligned} \tag{G1}$$

Taking twice the second expression minus the square of the first gives (I1). Since h is symmetric, $|e^h\omega|^2 = \omega^{\mathsf{T}} e^{2h}\omega$, proving the twist–tensor terms as well. The scalar coefficients in these expansions depend on r, L, η, V through (M3); expanding those coefficients adds higher-order scalar couplings.

Rotation invariance would allow $h_{ij}u_i v_j$ for two independent vectors u, v. Whether such an invariant occurs is decided by the actual action and the meaning of its variables. Here the mixing coordinates parametrize a coframe rotation and drop out of the scalar action. Their absence from (I1) is consequently an action-specific result. The derivative invariant Q in Section 4 belongs to the separately specified circle-dependent vector family.

H Numerical values and reproducible figures

The fixed illustrative point is $r = 3, L = \kappa = \eta = V = 1$, with inactive torsion coefficients set to zero for AX or VT. The analytic formulas give

$$\begin{aligned} &8\mathcal{K} = 24\pi^2, \qquad H_{\omega\omega} = \frac{2\pi^2}{3}, \qquad C_\omega = \frac{\pi^2}{3}, \\ &H_{\eta\eta} = H_{VV} = \frac{9\pi^2}{2}, \qquad H_{r\eta} = \frac{3\pi^2}{2}, \qquad H_{rV} = \frac{9\pi^2}{2} \quad \text{(MX)}, \\ &\epsilon_c = \frac{1}{\sqrt{2}} - 1 \simeq -0.292893218. \end{aligned} \tag{H1}$$

The `paper_results` package contains the static Cartan calculation and the restricted functions `shape_minimum`, `twist_bulk` and `scalar_bulk`. The `circle.joint_bulk` function implements (M4)–(M6), returning the static, velocity-quadratic and velocity-linear contributions separately. The `coupled_periodic_family` function evaluates (X4) by fixed periodic quadrature. Ten figure pairs include the cross coefficients and coupled sequence as well as the static and restricted-sector results. The data manifest records input parameters, grids, source hashes and data hashes.

From the repository's `script` directory, the commands are

```
python -B -m paper_results render -output-dir ../rendered
python -B -m paper_results recompute -output-dir ../recomputed-data
python -B -m paper_results render -data-dir ../recomputed-data
  -output-dir ../recomputed-figures
```

The last command is entered on one line. Output directories must be new. Rendering uses the saved arrays and does not invoke the geometric calculation. The notebook defaults to this saved-data route, while recalculation is explicitly selected. The attachments are

contained in the `data`, `LaTeX` and `script` directories. The entry point `script/README.md` links to dependencies, the viewer, and the convention and implementation guides in `script/docs`. The dataset index is `data/revised/manifest.json`; the figure notebook is `script/DPPUv2_Paper03_Figures_v3.ipynb`; the source is `LaTeX/main.tex`.

The repository is https://github.com/Muacca/DPPUv2-paper03, with the archival record at https://doi.org/10.5281/zenodo.19144481.

I Parameter dictionary and variation contract

The paper and `paper_results` use geometric torsion T. The code accompanying Papers01/02 uses the input convention $\tau = -T$, so its two torsion input parameters have the opposite signs. Tensor comparisons use this map; scalar expressions even in torsion do not detect it. The coefficient of the curvature side of complete NY is $1/2$, as in (I2). The circle has period one and the physical round radius is $r/2$ throughout.

On the static zero-twist axial slice, Paper02's coordinate is $x = a^{-2} = (1+\epsilon)^{-2/3}$ and its shape function satisfies $F(M_0^2) = -f(x)$. Paper01's round FULL potential is (S1); Paper02's potential reduces to (M3) on this slice at $\alpha = 0$. All three use the same geometric torsion and fixed-η radius variation, so the frame component $2\eta/r$ changes with r. The N used here is a scalar density; $N\,\mathrm{vol}_4$ is the NY four-form denoted N in Paper02.

The static formulas use constant frame components. The general circle action (M6) varies the same spatially homogeneous coframe through smooth periodic profiles. The derivative of M satisfies the determinant-one tangent condition, and the total-derivative boundary currents are integrated around the circle.

The shape and twist minima are statements about the model families. Scalar controls held fixed in one problem may be varied in another, but those problems must be identified. In particular, a global rotation of a static shape, a local coframe rotation, and a periodic coordinate transformation around the circle impose different equivalences.

J Joint circle action and velocity structure

J.1 Metric reduction and boundary current

Use the induced metric, lapse and shift (M4). The spatially homogeneous shift obeys (M5), and the extrinsic-curvature matrix is $K = (\gamma' - \ell)/(2\mathcal{N})$. The Euclidean curvature decomposition adds $[\mathrm{tr}(\gamma^{-1}K)]^2 - \mathrm{tr}[(\gamma^{-1}K)^2]$ to the intrinsic three-curvature, together with a divergence. Spatial divergences of invariant vectors vanish on this unimodular algebra. The remaining normal current is $-2\sqrt{\det\gamma}\,\mathrm{tr}(\gamma^{-1}K)$. Since $\mathrm{tr}Y = 0$ and $\mathcal{N}\sqrt{\det\gamma} = J$, it gives

$$\begin{aligned} R_{\mathrm{LC}} &= R_{\mathrm{stat,LC}} - \frac{\mathrm{tr}(X^2) - (\mathrm{tr}X)^2 - 2\mathrm{tr}(XY)}{4\mathcal{N}^2} + \frac{B_R'}{J}, \\ B_R &= -\frac{J}{\mathcal{N}^2}\mathrm{tr}X. \end{aligned} \tag{J1}$$

Here $R_{\mathrm{stat,LC}}$ is (A3) with $\eta = V = 0$. To see the subtraction explicitly, expand the quadratic form in $X - Y$: its Y^2 term is the zero-velocity contribution and combines with the intrinsic three-curvature into $R_{\mathrm{stat,LC}}$. Multiplication by $-\pi^2 J/(8\kappa^2)$ yields (M6) after the periodic boundary integral is removed. This derivation retains the non-Killing shift for anisotropic shape. The lapse–shift geometry is reviewed in [6]; the Euclidean sign here is also fixed directly by the Cartan curvature formula (A1).

J.2 Torsion derivatives and periodic rotations

For any smooth coframe rotation $O(y)$, let $E = E_3^y$ be the circle component of the final dual frame vector. Substituting the specified axial and trace torsions into the metric-compatible connection gives

$$\begin{aligned} R_{\mathrm{EC}} &= R_{\mathrm{LC}} - \frac{6\eta^2}{r^2} - \frac{2V^2}{3} + \frac{2}{J}\partial_y(JVE), \\ N_{\mathrm{NY}} &= -\frac{6}{J}\partial_y\left(J\frac{\eta}{r}E\right). \end{aligned} \tag{J2}$$

The quadratic torsion norms give the two algebraic terms; the trace-torsion divergence gives the last term in the first line. The axial NY current gives the second line. In the unrotated coframe, $E = 1/L$, so (J2) reduces to (A4)–(A5) at $\omega = 0$ and to (E3) at fixed scalars and round shape. Multiplication by the volume density makes both currents total derivatives, whose periodic integrals vanish. Thus η', V' and derivatives of O have been included; they do not create independent bulk velocity terms. No pointwise vanishing of the general NY density is required.

J.3 Cross terms and velocity rank

Expanding $M = e^h$ and (M4)–(M6) around a constant round, zero-twist point recovers the quadratic limit of (T6) and the representative cubic cross terms (X1). For the diagonal family, $\gamma = \mathrm{diag}(r^2e^{2s} + L^2w^2, r^2e^{-2s}, r^2)$; direct differentiation gives (X2) and (X3). For the rotating shape of (V6), γ' and ℓ are commutators with the same 01 rotation generator. Evaluating $-2\mathrm{tr}(XY)$ gives the displayed coefficient. These calculations retain the full doublet and vector components rather than inferring their couplings from a single static Hessian.

For the quadratic velocity form, $\gamma^{-1}\gamma'$ is similar to a symmetric matrix. Its tracefree part has five positive directions and its trace has one negative direction. Radius and five shape coordinates span every symmetric change of γ, so the Hessian in $(r, L, \eta, V, q_1, \ldots, q_5, \omega_0, \omega_1, \omega_2)$ has rank six, with inertia $(5, 1, 6)$, listing positive, negative and zero directions. At nonzero twist the null directions mix these coordinates. If all four scalar controls are fixed, the eight-dimensional shape–twist Hessian has inertia $(5, 0, 3)$ at zero twist and $(5, 1, 2)$ at nonzero twist: the twist variation then supplies the missing trace direction of the induced metric. This is an unreduced Euclidean velocity form. It does not count physical polarizations after lapse and momentum constraints. The term linear in velocities affects Euler equations while leaving this Hessian unchanged.

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